CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 03



By- Vishvadeep Gothi sir

Recap of Previous Lecture



Topic Pipeline Throughput

Topic Latency

Topic Instruction Pipeline

Topics to be Covered



- **Topic** Instruction Pipeline
- **Topic** Stall Cycles
- **Topic** Pipeline Hazard



Topic : Instruction Pipeline

If pipeline processing is implemented on Instruction Cycle



Topic : Instruction Pipeline

1. IF: Instruction Fetch
2. ID: Instruction Decode & Address Calculation
3. OF: Operand Fetch
4. EX: Execution
5. WB: Write Back



Topic : Instruction Pipeline

Clock Cycles

Instructions	1	2	3	4	5	6	7	8	9	10	11	12	13	14
I1	IF	ID	OF	EX	WB									
I2		IF	ID	OF	EX	WB								
I3			IF	ID	OF	EX	WB							
I4				IF	ID	OF	EX	WB						
I5					IF	-	-							
I6														
I7														
I8								IF	ID	OF	EX	WB		
I9									IF	ID	OF	EX	WB	

branch

Target

stalls or bubble



Topic : Instruction Pipeline

Assume in cycle 5, inst^n I4 decoded as branch type inst^n.
Then all inst^{ns} after I4 are removed from pipeline and CPU will not fetch any new inst^n until branch outcome is not available at the end of EX phase of I4.

Actually no. of inst^{ns} executed

$n = 6$ (I1 to I4 and I8 to I9)

$$\text{c/o any problem no. of cycles} = k + n - 1 = 5 + 6 - 1 = 10$$

But in example, 13 cycles are needed; because of 3 stall cycles due to branch inst^n.

If I4 will not take branch.

no. of instns executed (n) = 9

no. of cycles w/o any problems = 5 + 9 − 1 = 13

but 3 stalls also needed,

hence total cycles = 13 + 3 = 16

Assume that for branch inst^n, the ith stage provide target of branch (condition checked / branch evaluation done).

Then no. of stalls due to branch inst^n

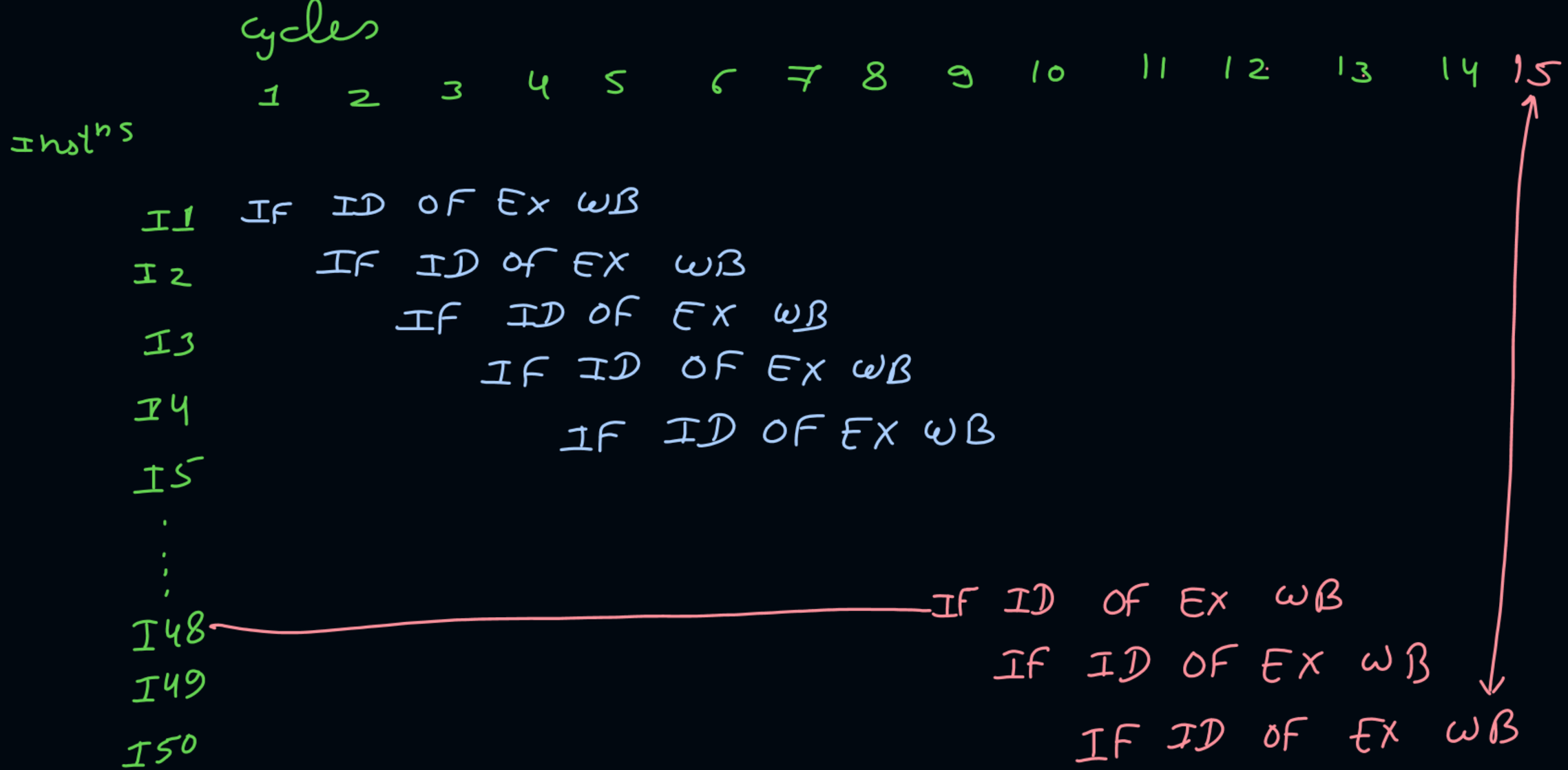
$= i$ — phase no. of inst^n fetch

$= \boxed{i - 1}$

If after branch inst^n, next inst^n is fetched after complete execution of branch inst^n then no. of stalls $= \boxed{k - 1}$

[NAT]

#Q. Consider a program which contains 50 instructions I1, I2 , I3 ..... I50. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I5 and its target is instruction I48. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is _____ ?


cycles
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Instns
I1
I2
I3
I4
I5
I48
I49
I50
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB
IF ID OF EX WB

shortcut :-

no. of inst^{ns} executed = 8 (I1 to I5 and I48 to I50)

w/o any hazard, no. of cycles = 5 + 8 − 1 = 12

stalls due to 1 branch inst^{n} = 4 − 1 = 3

Total = 15 cycles

[NAT]

#Q. In the above question if the pipeline has cycle time 10ns then total time required to execute the program is _____ ns?

$= 15 * 10$ ns

$= 150$ ns

[NAT]

#Q. Consider a program which contains 200 instructions I1, I2 , I3 ..... I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I6 and its target is instruction I16. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is _____ ?

I1 to I6 = 6 − 1 + 1 = 6

I16 to I200 = 200 − 16 + 1 = 185

Total (n) = 191

w/o hazard = 5 + 191 − 1 = 195

stalls due to branch = 4 − 1 = 3

Total = 198 Ans.

[NAT]

Ans = 207

#Q. Consider a program which contains 200 instructions I1, I2 , I3 ..... I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I8 and its target is instruction I108. If during the execution of the program the branch is not taken then number of cycles required to execute this program in the given pipeline is __?

$n = 200$ (because branch not taken)

$< 20\%$

w/o hazard = $5 + 200 - 1 = 204$

due to branch stalls = $4 - 1 = 3$

total = 207

Ques) In a prog. 4 branc instns.

Total no. of instns executed = 215

5-stage pipeline in which 3rd stage provide branch target.

1 cycle time = 15 ns

Prog. execution time = _____ ns

Soln

no. of cycles w/o hazard = 5+215-1 = 219

Total stalls = 4*(3-1) = 8

Total cycles = 227

Total time = 227 * 15 ns

= 3405 ns



Topic : Pipeline Hazards

Situations that prevent the next instruction from being executing during its designated clock cycle



Topic : Pipeline Hazards

1. Structural Hazard / Resource Conflict
2. Data Hazard / Data Dependency
3. Control Hazard / Branch Difficulty



Topic : Structural Hazard

→ always causes stalls until o/w not given in Quest^n.

2 different ~~segments~~ inst^{ns} try to use same resource at same time.

ex:- if MUL type inst^{ns} take 2 cycles in EX phase.

I1	MUL	IF	ID	OF	EX	EX	WB	
I2	ADD		IF	ID	OF	—	EX	WB

resource conflict

w/o hazard = 5 + 2 − 1 = 6

stall = 1

Total = 7



Topic : Data Hazard or Data Dependency

Result of an instruction is used as input in next

I1 :- $R1 \leftarrow R2 + R3$

I2 :- $R4 \leftarrow R1 * R5$

IF ID OF EX WB

IF ID – – OF EX WB

for I2, OF can be performed only after the WB of I1; so that correct input is fetched for I2.

example:-

```
R1 ← R2 + R3
R6 ← R5 * R4
R8 ← R1 / R9
```

1	2	3	4	5	6	7	8
IF	ID	OF	EX	WB			
	IF	ID	OF	EX	WB		
		IF	ID	—	OF	EX	WB

---

ex:-

```
   R1 ← R2 + R3
✓  R0 ← R5 − R4
✓  R9 ← R10 / R8
   R6 ← R1 − R7
```

IF	ID	OF	EX	WB			
	IF	ID	OF	EX	WB		
		IF	ID	OF	EX	WB	
			IF	ID	OF	EX	WB

solution of data hazard

software solution (by compiler)

⇓

→ Delayed Load

Hardware solution

⇓

→ Hardware Interlock

→ operand forwarding or bypassing

Delayed Load :-

Compiler either inserts independent instns or no-operation instns before dependent instn.

instn reshuffling
or
rescheduling

Hardware Interlock:-

providing stalls to solve data dependency.



2 mins Summary

Topic Pipeline Throughput

Topic Latency

Topic Instruction Pipeline



Happy Learning

THANK - YOU